

Mike Hosian Maeda

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EDUCATION

Alfred University

B.A. in Data Analytics & Computer Science – Double Major, GPA: 3.88/4.0

Alfred, NY

Expected May 2029

Relevant Coursework: Introduction to Computer Science, Data Structures & Algorithms, Object-Oriented Programming, Linear Algebra.

Honors: Alfred Startup Competition Winner (\$5,000), Presidential Scholarship, Honors Program, Dean's List '25 & '26, Omicron Delta Kappa Leadership Honor Society

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, SQL, C++, C, HTML/CSS

Frameworks & Libraries: Flask, React, REST APIs, PyTorch, TensorFlow, Scikit-learn, NumPy

Databases, Cloud & Tools: PostgreSQL, MySQL, SQLite, ChromaDB, Git, Linux

EXPERIENCE

Undergraduate Research Assistant

May 2026 – Present

Alfred University, Inamori School of Engineering | Python, TensorFlow

Alfred, NY

- Addressed a sustainability bottleneck in electronic-waste recycling, where slow manual sorting limits material recovery and drives landfill waste, by building a computer vision classification pipeline (TensorFlow, MobileNetV2) that identifies e-waste components across **28,000+ images** at **~90% validation accuracy**.
- Engineered an end-to-end Python inference pipeline integrating image preprocessing, model prediction, and database-backed recovery scoring, improving research processing throughput **50%**.
- Partnered with materials engineering researchers to translate recycling requirements into a reusable software workflow for automated component identification.

ITS Student HelpDesk Consultant

Sep 2025 – Present

Alfred University | Microsoft Entra ID, Identity & Access

Alfred, NY

- Selected among **13 student consultants** supporting production identity and access systems for **2,000+ users**, administering authentication and account provisioning through Microsoft Entra ID.
- Resolved **100+ hardware, software, and access incidents** monthly through root-cause troubleshooting, achieving near-complete first-contact resolution with minimal escalation and documenting solutions in the ITS ticketing system.
- Processed **10+ identity and access requests** daily, including MFA resets, account recovery, and authentication issues, rapidly restoring user access and reducing escalation to senior ITS staff.

Teaching Assistant, Analytics 101

Jan 2026 – Present

Alfred University | Excel, Data Analysis

Alfred, NY

- Facilitated Excel-based data analysis sessions for 30+ students, teaching structured formulas, what-if analysis, and financial modeling techniques applied to real business scenarios.

PROJECTS

School Management System | Java, Python, SQL, SQLite

[GitHub](#)

- Motivated by growing up in Tanzania, where many schools lack reliable internet and rely on slow, paper-based administration, engineered a fully offline-capable Java/Python records system managing **1,000+ entries** with object-oriented design and automated CRUD workflows, requiring no internet connectivity to operate.
- Designed a normalized SQL schema with enforced integrity constraints, improving administrative query efficiency **40%**.

Alfred RAG Assistant | Python, ChromaDB, Groq LLM, Gradio

[GitHub](#)

- Solved a real accessibility problem for Alfred students by building a retrieval-augmented generation (RAG) system in Python that answers student-life questions from a 10-document knowledge base, chunking content into **47 segments** and embedding with all-MiniLM-L6-v2 in a persistent ChromaDB vector store.
- Implemented a grounded generation pipeline (Groq Llama-3.3-70B, top-4 cosine retrieval) with a strict refusal mechanism and source-attributed responses to prevent hallucination, evaluated to **80% (4/5) response accuracy**.

Expense Tracker | Python, Flask, JavaScript, SQLite, REST APIs, OAuth, Vercel

[GitHub](#)

- Built to help students manage tight budgets without manual spreadsheets, engineered a full-stack web app (Flask backend, JavaScript/HTML/CSS frontend) with **20+ REST API endpoints** and a normalized SQLite schema, shipping Google OAuth, sessions, and budget tracking.
- Deployed to production on Vercel with automated email notifications, delivering **8+ user-facing features** across a scalable, cloud-hosted architecture.

LEADERSHIP AND ACTIVITIES

Resident Assistant, Alumni Relations & Outreach Chair @ ColorStack Alfred, Phonathon Ambassador

Alfred, NY

Alfred University

May 2026 – Present

- Selected as one of only a few freshmen to serve as Resident Assistant, driving community engagement for 30+ residents through peer mentorship and conflict resolution.
- Lead Alfred's ColorStack professional development and networking initiatives connecting underrepresented computing students with mentorship and career opportunities.
- Completed CodePath's AI 201 and Technical Interview Prep 103, and served as a Phonathon Ambassador engaging alumni donors.